Politecnico di Milano

Dipartimento di Elettronica, Informazione e Bioingegneria

prof. Fabrizio Ferrandi

Advanced Algorithms and Parallel Programming I part–January 26th, 2024

Polimi ID _____ _____

Surname _____ _____ **Name** _____

- This is a closed-book examination. You cannot use computers, phones, or laptops during the exam.
- Paper will be provided, but you should bring and use writing instruments that yield marks dark enough to be read easily. Erasable pens can be used.
- Total available time: 1h:00m.

Exercise 1 (4 points) _____

Exercise 2 (4 points) _____

Exercise 3 (4 points) _____

Exercise 4 (4 points) _____

Exercise n. 1

Answer the following questions about parallel application modeling and briefly explain (without an explanation, the answer will be considered invalid)

A. Please discuss the following sentence: "In a PRAM model, processors do not have private local memories". True/False? (1)

B. Please define what is meant by "Efficiency" in a PRAM application. (1)

C. Please provide an example of a PRAM application relying on a Common CW policy. (1)

D. Please relate the strong scaling vs the weak scaling to the Amdahl vs Gustafson laws. (1)

Exercise n. 2

Answer the following questions about algorithms and briefly explain (without an explanation, the answer will be considered invalid)

A. Please describe how the partitioning work in the "Select the *i* th smallest of *n* elements" algorithm. How is it randomized? (2)

B. What kind of time complexity has the longest common subsequence problem (1) **This question can be skipped in case of a passed challenge.**

C. Which is the type/class of randomized algorithms used by the primality testing algorithms presented during the lectures? (1)

Exercise n. 3

Answer the following questions about the advanced algorithm and briefly explain (without an explanation, the answer will be considered invalid)

A. Please discuss the amortized analysis. For which class of algorithm does it matter? (2)

B. Please provide the implementation of the split operation for Treaps. What is the operation complexity? (2)

Exercise n. 4

Answer the following questions about the algorithm analysis and briefly explain (without an explanation, the answer will be considered invalid)

A. Please describe the difference between on-line and off-line algorithms. How is their complexity analyzed? (2)

B. Please explain the differences between "Computed table" and "Unique table" with BDDs. Why are they used? (2)